HSC Mathematics Extension 1: Sequences and Series

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Download PDF and resources:
`https://github.com/vuhung16au/math-olympiad-ml`

"Sometimes it is the people no one imagines anything of who do the things that no one can imagine."

Alan Turing

Contents

1 Introduction

1.1 Project Overview

This booklet is a focused companion for NSW HSC Mathematics Extension 1 students studying sequences and series. It is designed to connect core sequence skills to exam-style problem solving with clear methods and structured solutions.

1.2 Target Audience

This resource is written for Year 12 students in Extension 1, and for tutors and teachers who want a compact booklet with consistent notation and worked examples.

1.3 How to Use This Booklet

- Read the fundamentals review first, then attempt the detailed problems in Part 1.
- Use Part 2 for timed practice and fluency, using hints only when needed.
- Keep a formula sheet while you work, but always justify each step from first principles.
- Track common slips, especially sign errors in geometric sums and indexing mistakes in sigma notation.

1.4 Topics Covered

This booklet develops the following Extension 1 ideas:

- defining sequences by term formula, recursion, and context
- arithmetic sequences and arithmetic series
- geometric sequences and geometric series
- finite sum formulas and sigma notation
- limiting sum of a geometric series when $|r| < 1$
- links between recurring decimals and geometric series
- mixed AP/GP problem solving in HSC-style questions

1.5 Where This Leads Next

Sequence fluency supports later work in calculus, proof, and Extension 2 style reasoning. In particular, convergence arguments and recursive definitions form a bridge to harder topics such as iterative methods and infinite processes.

2 Fundamentals Review

This section is a compact reference for notation, formulas, and common reasoning patterns used throughout the booklet.

2.1 How Sequences Are Specified

A sequence is an ordered list of numbers, usually written as $(a_n)_{n\geq 1}$. Common ways to define a sequence are:

- explicit formula: a_n written directly in terms of n
- recursive formula: each term written using previous terms
- contextual rule: terms generated from a worded process

Examples.

$$a_n = 3n - 1, \quad b_1 = 2,\ b_{n+1} = b_n + 5, \quad c_n = (-1)^n.$$

2.2 Arithmetic Sequences (AP)

An arithmetic sequence has constant difference d:

$$a_{n+1} - a_n = d.$$

If the first term is a, then

$$a_n = a + (n-1)d.$$

Problem 2.1: Recognizing an AP

Given $a_1 = 7$ and $a_{n+1} = a_n - 3$, find a_{10}.

Solution 2.1

Here $d = -3$, so

$$a_{10} = 7 + (10-1)(-3) = 7 - 27 = -20.$$

2.3 Arithmetic Series

The sum of the first n terms of an AP is:

$$S_n = \frac{n}{2}\big(2a + (n-1)d\big) = \frac{n}{2}(a + l),$$

where l is the nth term.

Problem 2.2: AP Sum

Find $\sum_{k=1}^{20}(5 + 2(k-1))$.

Solution 2.2

This is an AP with $a = 5$, $d = 2$, and $n = 20$. The 20th term is

$$l = 5 + 19 \cdot 2 = 43.$$

So

$$S_{20} = \frac{20}{2}(5 + 43) = 10 \cdot 48 = 480.$$

2.4 Geometric Sequences (GP)

A geometric sequence has constant ratio r:

$$\frac{a_{n+1}}{a_n} = r \quad (a_n \neq 0).$$

If the first term is a, then

$$a_n = ar^{n-1}.$$

Problem 2.3: Recognizing a GP

If $a_1 = 81$ and $r = \frac{1}{3}$, find a_5.

Solution 2.3

$$a_5 = 81\left(\frac{1}{3}\right)^4 = 81 \cdot \frac{1}{81} = 1.$$

2.5 Geometric Series

For $r \neq 1$, the sum of the first n terms is

$$S_n = a\frac{1 - r^n}{1 - r} = a\frac{r^n - 1}{r - 1}.$$

Use the form that keeps signs clean for your r value.

Problem 2.4: Finite GP Sum

Find $\sum_{k=0}^{6} 3 \cdot 2^k$.

Solution 2.4

This is $a = 3$, $r = 2$, $n = 7$ terms (from $k = 0$ to 6):

$$S_7 = 3\frac{2^7 - 1}{2 - 1} = 3(128 - 1) = 381.$$

2.6 Limiting Sum of a Geometric Series

If $|r| < 1$, then $r^n \to 0$ as $n \to \infty$, and the infinite sum exists:

$$S_\infty = \frac{a}{1 - r}.$$

If $|r| \geq 1$, the geometric series does not converge to a finite limiting sum.

Problem 2.5: Infinite GP

Evaluate $2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \cdots$.

Solution 2.5

Here $a = 2$, $r = \frac{1}{3}$, and $|r| < 1$, so

$$S_\infty = \frac{2}{1 - \frac{1}{3}} = \frac{2}{\frac{2}{3}} = 3.$$

2.7 Recurring Decimals and Geometric Series

Recurring decimals can be written as geometric series. For example,

$$0.\overline{7} = 0.7 + 0.07 + 0.007 + \cdots$$

is geometric with $a = 0.7$ and $r = 0.1$. So

$$0.\overline{7} = \frac{0.7}{1 - 0.1} = \frac{0.7}{0.9} = \frac{7}{9}.$$

The "Cyclic" Magic of $\frac{1}{7}$
The fraction $\frac{1}{7}$ produces the most famous repeating pattern in elementary number theory.
The Pattern:
$\frac{1}{7} = 0.\dot{1}4285\dot{7}$ (a 6-digit period).

2.8 Sequence Convergence Theorems

Monotone Convergence Theorem (MCT) The MCT provides a "shortcut" for proving a sequence converges by looking at its overall behavior rather than its specific terms.

- **The Rule:** If a sequence is monotone (always increasing or always decreasing) and it is bounded (it never goes past a certain value), then it must converge.

The Squeeze Theorem (Sandwich Theorem) The Squeeze Theorem is used when a sequence is "messy" (like one containing a sine or cosine function) but is trapped between two "cleaner" sequences that share the same limit.

- **The Rule:** Suppose we have three sequences (a_n), (b_n), and (c_n) such that for all n beyond a certain point:
$$a_n \leq b_n \leq c_n$$
- **The Result:** If $\lim_{n\to\infty} a_n = L$ and $\lim_{n\to\infty} c_n = L$, then the "squeezed" sequence in the middle must also have that limit: $\lim_{n\to\infty} b_n = L$.

2.9 Sigma and Pi Notation Quick Rules

Summation (Σ):

$$\sum_{k=1}^{n} (u_k + v_k) = \sum_{k=1}^{n} u_k + \sum_{k=1}^{n} v_k, \quad \sum_{k=1}^{n} cu_k = c \sum_{k=1}^{n} u_k.$$

Index shifts must preserve both the expression and bounds consistently.

Product (Π):

$$\prod_{k=1}^{n}(u_k v_k) = \left(\prod_{k=1}^{n} u_k\right)\left(\prod_{k=1}^{n} v_k\right), \quad \prod_{k=1}^{n} c u_k = c^n \prod_{k=1}^{n} u_k.$$

Note that a constant c is raised to the power of the number of terms when pulled outside a product, unlike sums where it acts as a simple multiplier.

Takeaways 2.1

- Always identify whether the question asks for a term a_n, a finite sum S_n, or a limit.
- AP questions depend on constant difference; GP questions depend on constant ratio.
- For infinite geometric sums, check $|r| < 1$ before using $S_\infty = \frac{a}{1-r}$.
- Track indexing carefully in sigma notation to avoid off-by-one errors.
- Chebyshev polynomial questions usually depend on either the recurrence relation or the trigonometric identities.

3 Part 1: Problems and Solutions (Detailed)

Part 1 contains full solutions with method explanations.

3.1 Basic Sequence Problems

Problem 3.1: Arithmetic nth term

An arithmetic sequence has first term 4 and common difference 5. Find a_{30}.

Solution 3.1

Use $a_n = a + (n-1)d$:

$$a_{30} = 4 + 29 \cdot 5 = 149.$$

Problem 3.2: Find the common ratio

In a geometric sequence, $a_2 = 12$ and $a_5 = 324$. Find the common ratio r and the first term a_1.

Solution 3.2

From $a_n = ar^{n-1}$,

$$a_2 = ar = 12, \qquad a_5 = ar^4 = 324.$$

Divide to remove a:

$$\frac{a_5}{a_2} = r^3 = \frac{324}{12} = 27 \implies r = 3.$$

Then $a_1 = a = \frac{12}{3} = 4$.

Problem 3.3: Sum of an AP

Find the sum of the first 40 terms of the arithmetic sequence

$$9,\ 13,\ 17,\ \ldots$$

Solution 3.3

Here $a = 9$, $d = 4$, $n = 40$. The 40th term is

$$l = 9 + 39 \cdot 4 = 165.$$

So

$$S_{40} = \frac{40}{2}(9 + 165) = 20 \cdot 174 = 3480.$$

Problem 3.4: Recurring decimal as a fraction

Express $0.\overline{23}$ as a rational number.

Solution 3.4

$$0.\overline{23} = \frac{23}{99}.$$

A geometric view is

$$0.23 + 0.0023 + 0.000023 + \cdots$$

with first term $\frac{23}{100}$ and ratio $\frac{1}{100}$.

Problem 3.5: Chebyshev recurrence: first kind

The Chebyshev polynomials of the first kind satisfy

$$T_0(x) = 1, \qquad T_1(x) = x, \qquad T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

Use the recurrence to find $T_2(x)$ and $T_3(x)$, then evaluate $T_3\left(\frac{1}{2}\right)$.

Hint: Build the polynomials one at a time: first use $T_2 = 2xT_1 - T_0$, then use $T_3 = 2xT_2 - T_1$.

Solution 3.5

$$T_2(x) = 2x^2 - 1.$$

Then

$$T_3(x) = 2x(2x^2 - 1) - x = 4x^3 - 3x.$$

So

$$T_3\left(\frac{1}{2}\right) = 4\left(\frac{1}{8}\right) - 3\left(\frac{1}{2}\right) = \frac{1}{2} - \frac{3}{2} = -1.$$

Takeaways 3.1

- Chebyshev polynomials can be generated recursively from two starting values.
- Work in order; each new polynomial depends on the previous two.
- Substitution is easiest after simplifying the polynomial.

3.2 Medium Sequence Problems

Problem 3.6: Mixed AP constraints

An arithmetic sequence has $a_3 = 11$ and $a_{12} = 47$. Find a_1 and d, then determine S_{25}.

Solution 3.6

Use $a_n = a + (n-1)d$:

$$a + 2d = 11, \qquad a + 11d = 47.$$

Subtract:

$$9d = 36 \implies d = 4,$$

then $a = 11 - 8 = 3$. Now

$$a_{25} = 3 + 24 \cdot 4 = 99,$$

so

$$S_{25} = \frac{25}{2}(3 + 99) = \frac{25}{2} \cdot 102 = 1275.$$

Problem 3.7: Finite GP with unknown index

For the geometric sequence with first term 5 and ratio 2, find n such that

$$S_n = 2555.$$

Solution 3.7

$$S_n = 5\frac{2^n - 1}{2 - 1} = 5(2^n - 1) = 2555.$$

So

$$2^n - 1 = 511 \implies 2^n = 512 = 2^9.$$

Hence $n = 9$.

Problem 3.8: Financial sequences and geometric sums

A loan of $\$L$ is taken out at an interest rate of r per period. At the end of each period, a constant repayment of $\$P$ is made after interest has been added to the balance. Let M_n be the balance owing after n periods.

(i) Write down expressions for M_1 and M_2 in terms of L, r, and P.

(ii) Show that the balance after n periods can be written as

$$M_n = L(1+r)^n - P\left[1 + (1+r) + (1+r)^2 + \cdots + (1+r)^{n-1}\right].$$

(iii) Using the formula for the sum of a geometric progression, prove that

$$M_n = L(1+r)^n - \frac{P((1+r)^n - 1)}{r}.$$

(iv) A car loan of \$20 000 is taken at 1% interest per month. Calculate the monthly repayment P required to reduce the balance to zero at the end of 24 months.

Hint:

- **For (i):** Add interest first, then subtract the repayment.
- **For (ii):** Expand the first few balances and look for the pattern.
- **For (iii):** The bracket is a finite GP with first term 1 and ratio $1 + r$.
- **For (iv):** Set $M_{24} = 0$, with $L = 20000$ and $r = 0.01$.

Solution 3.8

(i) After one period,

$$M_1 = L(1+r) - P.$$

After two periods,

$$M_2 = (L(1+r) - P)(1+r) - P = L(1+r)^2 - P(1+r) - P.$$

(ii) Continuing this pattern gives

$$M_n = L(1+r)^n - P\left[(1+r)^{n-1} + (1+r)^{n-2} + \cdots + (1+r) + 1\right].$$

Reordering the bracket,

$$M_n = L(1+r)^n - P\left[1 + (1+r) + (1+r)^2 + \cdots + (1+r)^{n-1}\right].$$

(iii) The bracket is a finite GP with first term 1, ratio $1+r$, and n terms:

$$1 + (1+r) + \cdots + (1+r)^{n-1} = \frac{(1+r)^n - 1}{(1+r) - 1} = \frac{(1+r)^n - 1}{r}.$$

Hence

$$M_n = L(1+r)^n - \frac{P((1+r)^n - 1)}{r}.$$

(iv) Here $L = 20000$, $r = 0.01$, and $n = 24$. Set $M_{24} = 0$:

$$0 = 20000(1.01)^{24} - \frac{P((1.01)^{24} - 1)}{0.01}.$$

Thus

$$P = \frac{20000(1.01)^{24}(0.01)}{(1.01)^{24} - 1}.$$

Using $(1.01)^{24} \approx 1.269735$,

$$P \approx \$941.47.$$

Takeaways 3.2

- Loan balances can be modelled by applying interest and repayment recursively.
- Repeated repayments form a finite geometric sum after compounding.
- Setting the final balance to zero gives the repayment required to clear a loan.

Problem 3.9: Telescoping sums and inductive proof

Consider the sequence

$$a_k = \frac{2}{k(k+1)(k+2)} \qquad (k \geq 1).$$

Define the partial sums

$$S_n = \sum_{k=1}^{n} a_k.$$

(i) Show, using partial fractions, that

$$\frac{2}{k(k+1)(k+2)} = \frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)}.$$

(ii) Demonstrate that the sequence of partial sums telescopes. Hence find a simplified expression for S_n in terms of n.

(iii) Prove the expression for S_n found in part (ii) by mathematical induction for all $n \geq 1$.

(iv) Determine the limiting sum of the series as $n \to \infty$.

Hint:

- **For (i):** Put the two fractions over the common denominator $k(k+1)(k+2)$.
- **For (ii):** Write out the first few terms of S_n and look for cancellation.
- **For (iii):** Use $S_{m+1} = S_m + a_{m+1}$ in the induction step.
- **For (iv):** Take the limit of the closed form for S_n.

Solution 3.9

(i)

$$\frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)} = \frac{k+2-k}{k(k+1)(k+2)} = \frac{2}{k(k+1)(k+2)}.$$

(ii) Therefore

$$a_k = \frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)}.$$

So

$$\begin{aligned} S_n &= \left(\frac{1}{1\cdot 2} - \frac{1}{2\cdot 3}\right) + \left(\frac{1}{2\cdot 3} - \frac{1}{3\cdot 4}\right) + \cdots + \left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right) \\ &= \frac{1}{2} - \frac{1}{(n+1)(n+2)}. \end{aligned}$$

(iii) We prove

$$S_n = \frac{1}{2} - \frac{1}{(n+1)(n+2)}.$$

For $n = 1$,

$$S_1 = \frac{2}{1\cdot 2\cdot 3} = \frac{1}{3}$$

and

$$\frac{1}{2} - \frac{1}{2\cdot 3} = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}.$$

Now assume the formula holds for $n = m$. Then

$$\begin{aligned} S_{m+1} &= S_m + \frac{2}{(m+1)(m+2)(m+3)} \\ &= \frac{1}{2} - \frac{1}{(m+1)(m+2)} + \left(\frac{1}{(m+1)(m+2)} - \frac{1}{(m+2)(m+3)}\right) \\ &= \frac{1}{2} - \frac{1}{(m+2)(m+3)}. \end{aligned}$$

This is the required formula with $n = m+1$, so the result holds for all $n \geq 1$.

(iv)

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \left(\frac{1}{2} - \frac{1}{(n+1)(n+2)}\right) = \frac{1}{2}.$$

Takeaways 3.3

- Telescoping sums work by rewriting terms so that most neighbouring parts cancel.
- Induction is a useful way to verify a closed form found by pattern spotting.
- A finite partial-sum formula can make the limiting sum immediate.

Problem 3.10: Binomial coefficients and partial sums

Let

$$S_n = \sum_{k=0}^{n} \binom{n}{k}$$

be the sum of the binomial coefficients for a fixed integer $n \geq 1$.

(i) Use the binomial theorem expansion of $(1+x)^n$ to prove that $S_n = 2^n$.

(ii) Use Pascal's identity

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

to prove the recurrence relation $S_n = 2S_{n-1}$.

(iii) Evaluate the alternating partial sum

$$A_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k}$$

and explain why the sum of even-indexed coefficients equals the sum of odd-indexed coefficients.

(iv) By differentiating the expansion of $(1+x)^n$ with respect to x, show that

$$\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}.$$

Hint:

- **For (i):** Substitute $x = 1$ into the binomial theorem.
- **For (ii):** Sum Pascal's identity over the appropriate range of k.
- **For (iii):** Substitute $x = -1$ into the binomial theorem.
- **For (iv):** Differentiate first, then substitute $x = 1$.

Solution 3.10

(i) By the binomial theorem,

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

Putting $x=1$ gives

$$2^n = \sum_{k=0}^{n} \binom{n}{k} = S_n.$$

(ii) Using Pascal's identity,

$$\begin{aligned} S_n &= \sum_{k=0}^{n} \binom{n}{k} \\ &= \sum_{k=0}^{n} \binom{n-1}{k} + \sum_{k=0}^{n} \binom{n-1}{k-1}. \end{aligned}$$

The out-of-range terms are zero, so each sum equals S_{n-1}. Hence

$$S_n = 2S_{n-1}.$$

(iii) By the binomial theorem again,

$$A_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} = (1-1)^n = 0.$$

Therefore

$$\sum_{\substack{0 \le k \le n \\ k \text{ even}}} \binom{n}{k} - \sum_{\substack{0 \le k \le n \\ k \text{ odd}}} \binom{n}{k} = 0,$$

so the even-indexed and odd-indexed sums are equal.

(iv) Differentiate

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$$

to get

$$n(1+x)^{n-1} = \sum_{k=1}^{n} k \binom{n}{k} x^{k-1}.$$

Putting $x=1$ gives

$$\sum_{k=1}^{n} k \binom{n}{k} = n2^{n-1}.$$

Takeaways 3.4

- Substituting special values into the binomial theorem turns an expansion into a sum identity.
- Pascal's identity explains why the total sum of coefficients doubles from one row to the next.
- Differentiating a generating expansion is a powerful way to evaluate weighted sums.

Problem 3.11: The Recursive Geometry of Chebyshev Sequences

Let a sequence of functions $(T_n(x))$ be defined for $n \geq 0$ by the recurrence relation

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x),$$

with initial terms $T_0(x) = 1$ and $T_1(x) = x$.

(i) Calculate the explicit polynomial expressions for $T_2(x)$ and $T_3(x)$.

(ii) Using the identity

$$\cos(A + B) + \cos(A - B) = 2 \cos A \cos B,$$

prove by mathematical induction that for all $n \geq 0$,

$$T_n(\cos\theta) = \cos(n\theta).$$

(iii) Let x_0 be a root of $T_n(x)$. Show that $|x_0| \leq 1$. Hence, determine the exact values of the n distinct roots of $T_n(x)$ in terms of n.

(iv) Consider the sequence of values $v_n = T_n(1.5)$. Show that (v_n) satisfies a linear recurrence relation with constant coefficients. Find a closed-form expression for v_n and determine the value of the limit

$$\lim_{n\to\infty} \frac{v_{n+1}}{v_n}.$$

Hint:

- **For (i):** Substitute the previous two polynomials into the recurrence formula.
- **For (ii):** Use the sum-to-product identity with $A = k\theta$ and $B = \theta$.
- **For (iii):** Solve $\cos(n\theta) = 0$ for θ, then use $x = \cos\theta$.
- **For (iv):** Solve the characteristic equation for the recurrence $v_{n+1} = 3v_n - v_{n-1}$.

Solution 3.11

(i) Directly from the recurrence,

$$T_2(x) = 2x^2 - 1, \qquad T_3(x) = 2x(2x^2 - 1) - x = 4x^3 - 3x.$$

(ii) The cases $n = 0, 1$ give $1 = \cos 0$ and $T_1(\cos\theta) = \cos\theta$. Suppose

$$T_k(\cos\theta) = \cos(k\theta), \qquad T_{k-1}(\cos\theta) = \cos((k-1)\theta).$$

Then

$$\begin{aligned} T_{k+1}(\cos\theta) &= 2\cos\theta\, T_k(\cos\theta) - T_{k-1}(\cos\theta) \\ &= 2\cos\theta\cos(k\theta) - \cos((k-1)\theta) \\ &= \cos((k+1)\theta), \end{aligned}$$

using $\cos((k+1)\theta) + \cos((k-1)\theta) = 2\cos(k\theta)\cos\theta$. Hence $T_n(\cos\theta) = \cos(n\theta)$ for all $n \geq 0$.

(iii) If $x > 1$, write $x = \cosh u$; then $T_n(x) = \cosh(nu) > 0$. If $x < -1$, write $x = -\cosh u$; then $T_n(x) = (-1)^n \cosh(nu) \neq 0$. Thus every real root satisfies $|x_0| \leq 1$.

Now write $x = \cos\theta$. The roots satisfy

$$T_n(\cos\theta) = 0 \iff \cos(n\theta) = 0 \iff n\theta = \frac{(2k-1)\pi}{2}.$$

So the n distinct roots are

$$x_k = \cos\left(\frac{(2k-1)\pi}{2n}\right), \qquad k = 1, 2, \ldots, n.$$

(iv) With $v_n = T_n(1.5)$,

$$v_{n+1} = 3v_n - v_{n-1}, \qquad v_0 = 1, \quad v_1 = \frac{3}{2}.$$

The characteristic equation is $\lambda^2 - 3\lambda + 1 = 0$, with roots

$$\alpha = \frac{3+\sqrt{5}}{2}, \qquad \beta = \frac{3-\sqrt{5}}{2}.$$

Hence $v_n = A\alpha^n + B\beta^n$. From $v_0 = 1$ and $v_1 = \frac{3}{2}$, $A = B = \frac{1}{2}$, so

$$v_n = \frac{1}{2}\alpha^n + \frac{1}{2}\beta^n.$$

Since $|\beta| < \alpha$, the dominant term is α^n, and therefore

$$\lim_{n\to\infty} \frac{v_{n+1}}{v_n} = \alpha = \frac{3+\sqrt{5}}{2}.$$

Takeaways 3.5

- The recurrence produces the Chebyshev polynomials one term at a time.
- The identity $T_n(\cos\theta) = \cos(n\theta)$ turns polynomial roots into trigonometric roots.
- Evaluating a polynomial recurrence at a fixed value gives an ordinary numerical recurrence.

3.3 Advanced Sequence Problems

Problem 3.12: Recursive sequence and limiting value

Let (x_n) be defined by

$$x_1 = 16, \qquad x_{n+1} = 2 + \frac{1}{x_n} \quad (n \geq 1).$$

(i) Find exact values of x_2, x_3, x_4.

(ii) Assuming $x_n \to L > 0$, show that $L = 1 + \sqrt{2}$.

(iii) Prove that

$$|x_{n+1} - L| = \frac{|x_n - L|}{Lx_n},$$

and deduce that for $n \geq 2$,

$$|x_{n+1} - L| < \frac{1}{4}|x_n - L|.$$

(iv) Hence show $x_n \to 1 + \sqrt{2}$.

Solution 3.12

(i)

$$x_2 = 2 + \frac{1}{16} = \frac{33}{16}, \quad x_3 = 2 + \frac{16}{33} = \frac{82}{33}, \quad x_4 = 2 + \frac{33}{82} = \frac{197}{82}.$$

For $n \geq 2$, we have $x_n > 2$ because each term is $2 + \frac{1}{x_{n-1}}$ with positive second part.
(ii) If $x_n \to L$, then taking limits in the recurrence gives

$$L = 2 + \frac{1}{L} \implies L^2 - 2L - 1 = 0.$$

So $L = 1 \pm \sqrt{2}$. Since terms are positive and > 2 from $n \geq 2$, we take

$$L = 1 + \sqrt{2}.$$

(iii)

$$x_{n+1} - L = \left(2 + \frac{1}{x_n}\right) - \left(2 + \frac{1}{L}\right) = \frac{L - x_n}{Lx_n},$$

thus

$$|x_{n+1} - L| = \frac{|x_n - L|}{Lx_n}.$$

Now $L = 1 + \sqrt{2} > 2$ and $x_n > 2$ for $n \geq 2$, so $Lx_n > 4$. Hence

$$|x_{n+1} - L| < \frac{1}{4}|x_n - L|, \quad n \geq 2.$$

(iv) Repeatedly applying the inequality gives geometric decay of the error:

$$|x_n - L| \leq \left(\frac{1}{4}\right)^{n-2} |x_2 - L| \to 0.$$

Therefore $x_n \to L = 1 + \sqrt{2}$.

Takeaways 3.6

- For recursive limits, first solve the fixed-point equation.
- A contraction inequality gives a rigorous convergence proof.
- The same method works for many rational recursions.
- The sequence x_n is related to the continued fraction of $1+\sqrt{2}$. It can be expressed as

$$x_{n+1} = 2 + \frac{1}{x_n} = 2 + \frac{1}{2 + \frac{1}{x_{n-1}}} = \cdots = 2 + \frac{1}{2 + \frac{1}{2+\cdots+\frac{1}{x_1}}}$$

Problem 3.13: The Dynamics of Iteration

Consider the function

$$g(x) = \sqrt{x+2}.$$

We define an iterative sequence by

$$x_{n+1} = g(x_n), \qquad x_1 = 1.$$

(i) Calculate x_2, x_3, and x_4 to three decimal places. By solving

$$L = \sqrt{L+2},$$

determine the exact positive limit L that the sequence appears to be approaching.

(ii) Show that

$$0 < g'(x) < \frac{1}{2}$$

for all $x > 0$.

(iii) Prove that

$$|x_{n+1} - L| < \frac{1}{2}|x_n - L|.$$

Hence deduce that the sequence converges to L for any initial value $x_1 > 0$.

(iv) Define a function $f(x)$ such that Newton's method would converge to the same limit L found in part (i), and write down the specific iterative formula for this $f(x)$.

Hint:

- **For (i):** Substitute repeatedly into $x_{n+1} = \sqrt{x_n + 2}$, then solve the fixed-point equation.
- **For (ii):** Differentiate $g(x) = \sqrt{x+2}$ and note where the derivative is largest for $x > 0$.
- **For (iii):** Use either the mean value theorem or rationalise $\sqrt{x_n+2} - \sqrt{L+2}$.
- **For (iv):** Rearrange $L = \sqrt{L+2}$ into a polynomial equation.

Solution 3.13

(i)

$$x_2 = \sqrt{3} \approx 1.732, \quad x_3 = \sqrt{2+\sqrt{3}} \approx 1.932, \quad x_4 \approx 1.983.$$

If $x_n \to L > 0$, then

$$L = \sqrt{L+2} \implies L^2 - L - 2 = 0 \implies (L-2)(L+1) = 0.$$

Thus the positive limit is

$$L = 2.$$

(ii)

$$g'(x) = \frac{1}{2\sqrt{x+2}}.$$

For $x > 0$, this is positive and

$$g'(x) < \frac{1}{2\sqrt{2}} < \frac{1}{2}.$$

(iii) Since $L = 2$ and $g(L) = L$,

$$\begin{aligned} |x_{n+1} - L| &= \left|\sqrt{x_n+2} - \sqrt{L+2}\right| \\ &= \frac{|x_n - L|}{\sqrt{x_n+2} + \sqrt{L+2}} \\ &< \frac{1}{2}|x_n - L|, \end{aligned}$$

because $x_n > 0$ and $\sqrt{L+2} = 2$. Repeating this inequality gives

$$|x_n - L| < \left(\frac{1}{2}\right)^{n-1} |x_1 - L| \to 0,$$

so $x_n \to L = 2$ for any $x_1 > 0$.
(iv) From the fixed-point equation,

$$L^2 - L - 2 = 0.$$

So take

$$f(x) = x^2 - x - 2.$$

Newton's method gives

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - x_n - 2}{2x_n - 1}.$$

Takeaways 3.7

- Fixed point is the value of x such that $f(x) = x$.
- Fixed points often reveal the likely limit of an iterative sequence.
- A contraction inequality proves convergence by forcing errors to shrink geometrically.
- Newton's method can target the same limit by solving the fixed-point equation as a root problem.

Problem 3.14: The Geometry of Harmonic Means

A sequence (h_n) is in harmonic progression (HP) if the sequence of reciprocals

$$x_n = \frac{1}{h_n}$$

forms an arithmetic progression (AP).

(i) Given that the third term of an HP is 1 and the sixth term is $\frac{1}{2}$, find the first term h_1 and the common difference d of the underlying AP.

(ii) The harmonic mean H of two positive numbers a and b is defined so that a, H, b is an HP. Show algebraically that

$$H = \frac{2ab}{a+b}.$$

(iii) Let

$$A = \frac{a+b}{2}, \qquad G = \sqrt{ab}$$

be the arithmetic and geometric means of a and b. Prove that $G^2 = AH$. Hence prove that, for distinct positive real numbers a and b,

$$H < G < A.$$

(iv) Consider the infinite harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n}.$$

By comparing the sum of the first 2^k terms to blocks of constant lower bounds, prove that this harmonic series does not have a finite limiting sum.

Hint:

- **For (i):** Convert the HP terms into AP terms x_3 and x_6, then use $x_n = a + (n-1)d$.
- **For (ii):** If a, H, b is an HP, then $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ is an AP.
- **For (iii):** Multiply A and H. For the inequality, use $(a-b)^2 > 0$ or $(\sqrt{a} - \sqrt{b})^2 > 0$.
- **For (iv):** Group terms as $\left(\frac{1}{3} + \frac{1}{4}\right)$, then $\left(\frac{1}{5} + \cdots + \frac{1}{8}\right)$, and so on.

Solution 3.14

(i) Since $x_n = \frac{1}{h_n}$,

$$x_3 = 1, \qquad x_6 = 2.$$

For the underlying AP,

$$x_6 - x_3 = 3d \implies 2 - 1 = 3d \implies d = \frac{1}{3}.$$

Also

$$x_1 = x_3 - 2d = 1 - \frac{2}{3} = \frac{1}{3},$$

so

$$h_1 = \frac{1}{x_1} = 3.$$

(ii) Since $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ is an AP, the middle term is the average of the outer terms:

$$\frac{1}{H} = \frac{1}{2}\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{a+b}{2ab}.$$

Therefore

$$H = \frac{2ab}{a+b}.$$

(iii) Using the formulas for A and H,

$$AH = \frac{a+b}{2} \cdot \frac{2ab}{a+b} = ab = G^2.$$

For $a \neq b$,

$$(\sqrt{a} - \sqrt{b})^2 > 0 \implies a + b > 2\sqrt{ab} \implies A > G.$$

Since $G^2 = AH$ and all quantities are positive, $A > G$ gives

$$H = \frac{G^2}{A} < G.$$

Hence

$$H < G < A.$$

(iv) Group the harmonic series into dyadic blocks:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots.$$

Each block after the first has sum greater than or equal to $\frac{1}{2}$:

$$\frac{1}{3} + \frac{1}{4} > \frac{1}{4} + \frac{1}{4} = \frac{1}{2}, \qquad \frac{1}{5} + \cdots + \frac{1}{8} > 4 \cdot \frac{1}{8} = \frac{1}{2},$$

and similarly

$$\frac{1}{2^{j-1}+1} + \cdots + \frac{1}{2^j} > 2^{j-1} \cdot \frac{1}{2^j} = \frac{1}{2}.$$

Thus the partial sums exceed

$$1 + \frac{1}{2} + \underbrace{\frac{1}{2} + \frac{1}{2} + \cdots + \frac{1}{2}}_{\text{arbitrarily many terms}},$$

so they grow without bound. Therefore the harmonic series has no finite limiting sum.

Takeaways 3.8

- Harmonic progressions are arithmetic progressions viewed through reciprocals.
- For positive distinct numbers, the classical means satisfy $H < G < A$.
- A sequence can have terms tending to zero while its infinite series still diverges.

Problem 3.15: Chebyshev composition

The first-kind Chebyshev polynomials $T_n(x)$ are defined by the recurrence

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

It is given that $T_n(\cos\theta) = \cos(n\theta)$ for all $n \geq 0$ and $\theta \in \mathbb{R}$. (Do NOT prove this fact.) For first-kind Chebyshev polynomials, prove that

$$T_m(T_n(x)) = T_{mn}(x)$$

for all x of the form $x = \cos\theta$. Then use this result to find

$$T_2(T_3(x)).$$

Hint: Use $T_n(\cos\theta) = \cos(n\theta)$. For the explicit polynomial, use $T_6(x)$.

Solution 3.15

Let $x = \cos\theta$. Then

$$T_m(T_n(x)) = T_m(\cos(n\theta)) = \cos(mn\theta) = T_{mn}(\cos\theta) = T_{mn}(x).$$

Therefore

$$T_2(T_3(x)) = T_6(x).$$

Using the recurrence after $T_5(x) = 16x^5 - 20x^3 + 5x$,

$$T_6(x) = 2xT_5(x) - T_4(x) = 32x^6 - 48x^4 + 18x^2 - 1.$$

Takeaways 3.9

- The trigonometric definition makes Chebyshev composition simple.
- Composition of first-kind Chebyshev polynomials multiplies the indices.
- Hard-looking polynomial composition can often be avoided by using structure.

4 Part 2: Problems with Hints and Solutions (Concise)

Part 2 gives extra practice with upside-down hints and concise solutions.

4.1 Basic Sequence Problems

Problem 4.1: Quick AP term

Find a_{18} for the arithmetic sequence $2, 7, 12, \ldots$.

Hint: Use $a_n = a + (n-1)d$ with $a = 2$ and $d = 5$.

Solution 4.1

$$a_{18} = 2 + 17 \cdot 5 = 87.$$

Problem 4.2: Quick GP sum

Find

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{128}.$$

Hint: This is a finite GP with $a = 1$, $r = \frac{1}{2}$, and 8 terms.

Solution 4.2

$$S_8 = \frac{1 - (\frac{1}{2})^8}{1 - \frac{1}{2}} = 2\left(1 - \frac{1}{256}\right) = \frac{255}{128}.$$

Problem 4.3: Quick Chebyshev value

For the second-kind Chebyshev polynomial

$$U_2(x) = 4x^2 - 1,$$

find $U_2\left(\frac{1}{2}\right)$.

Hint: Substitute $x = \frac{1}{2}$ carefully.

Solution 4.3

$$U_2\left(\frac{1}{2}\right) = 4\left(\frac{1}{4}\right) - 1 = 1 - 1 = 0.$$

Takeaways 4.1

- For a given Chebyshev polynomial, direct substitution is often enough.
- Square fractions before multiplying by the coefficient.

Problem 4.4: Arithmetic pricing model

An installer charges \$400 for the first window and \$100 for each additional window.

(i) Write the total cost for 1, 2, 3, and 4 windows.

(ii) Find a formula for the total cost A_n for n windows.

(iii) Find A_{15}.

(iv) If the budget is strictly less than \$10,000, what is the maximum number of windows?

Hint: Model A_n as an arithmetic sequence with first term 400 and difference 100.

Solution 4.4

$$A_n = 400 + (n-1)100 = 100n + 300.$$

So $A_{15} = 1800$. For the budget:

$$100n + 300 < 10000 \implies n < 97,$$

hence the maximum integer is $n = 96$.

Problem 4.5: Simple interest and AP

A principal of \$2,000,000 is invested at 5.7% p.a. simple interest. Let A_n be the total amount after n years.

(i) Find A_1, A_2, A_3, A_4.

(ii) Find a formula for A_n and evaluate A_{12}.

(iii) How many whole years before the amount exceeds \$6,000,000?

Hint: Simple interest adds a fixed amount each year, so (A_n) is arithmetic.

Solution 4.5

Yearly interest is $0.057 \times 2{,}000{,}000 = 114{,}000$. Hence

$$A_n = 2{,}000{,}000 + 114{,}000n.$$

So

$$A_1 = 2{,}114{,}000,\ A_2 = 2{,}228{,}000,\ A_3 = 2{,}342{,}000,\ A_4 = 2{,}456{,}000,$$

and

$$A_{12} = 3{,}368{,}000.$$

For exceeding \$6,000,000:

$$2{,}000{,}000 + 114{,}000n > 6{,}000{,}000 \implies n > 35.087\ldots$$

so $n = 36$ years.

Problem 4.6: Recurring decimal to fraction

Convert each to a fraction:

$$0.2, \qquad 0.\dot{2}, \qquad 0.1\dot{2}.$$

Hint: Use place value for terminating decimals and GP ideas for recurring decimals.

Solution 4.6

$$0.2 = \frac{1}{5}, \qquad 0.\dot{2} = \frac{2}{9}, \qquad 0.1\dot{2} = \frac{11}{90}.$$

Problem 4.7: Nested shaded squares

In a unit square, shade the bottom-left quarter. Then repeatedly subdivide the top-right quarter into four equal squares and shade its bottom-left quarter. Find the total shaded fraction.

Hint: Shaded areas form an infinite GP with first term $\frac{1}{4}$ and ratio $\frac{1}{4}$.

Solution 4.7

$$S_\infty = \frac{\frac{1}{4}}{1-\frac{1}{4}} = \frac{1}{3}.$$

So the shaded fraction is $\frac{1}{3}$.

Problem 4.8: Fraction of a recurring decimal

Find the fraction for $0.\dot{1}2\dot{3}$?

Hint: Use $a = \frac{123}{1000}$ and $r = \frac{1}{1000}$ in your S_∞ formula.

Solution 4.8

$$S_\infty = \frac{\frac{123}{1000}}{1 - \frac{1}{1000}} = \frac{123}{999} = \frac{41}{333}.$$

Takeaways 4.2

- Every recurring decimal is a geometric series sum S_∞.

4.2 Medium Sequence Problems

Problem 4.9: Indexing with sigma notation

Given

$$\sum_{k=1}^{n}(3k-1) = 119,$$

find n.

Hint: Split the sum into $3\sum k - \sum 1$.

Solution 4.9

$$\sum_{k=1}^{n}(3k-1) = 3 \cdot \frac{n(n+1)}{2} - n = \frac{3n^2+n}{2}.$$

Set equal to 119:

$$\frac{3n^2+n}{2} = 119 \implies 3n^2 + n - 238 = 0.$$

$$\Delta = 1 + 2856 = 2857.$$

This is not a perfect square, so there is no integer n with exact value 119.

Problem 4.10: Infinite geometric interpretation

Evaluate

$$0.4\overline{2} = 0.42222\ldots$$

as a fraction.

Hint: Write as $0.4 + 0.02222\ldots$ and convert the recurring part using a GP.

Solution 4.10

$$0.02222\ldots = \frac{2}{90} = \frac{1}{45}.$$

So

$$0.4\overline{2} = \frac{2}{5} + \frac{1}{45} = \frac{18}{45} + \frac{1}{45} = \frac{19}{45}.$$

Problem 4.11: Second-kind Chebyshev recurrence

The second-kind Chebyshev polynomials satisfy

$$U_0(x) = 1, \qquad U_1(x) = 2x, \qquad U_{n+1}(x) = 2xU_n(x) - U_{n-1}(x).$$

Use this recurrence to find $U_3(x)$ and $U_4(x)$.

Hint: First find $U_2(x)$, then continue to $U_3(x)$ and $U_4(x)$.

Solution 4.11

$$U_2(x) = 2x(2x) - 1 = 4x^2 - 1.$$
$$U_3(x) = 2x(4x^2 - 1) - 2x = 8x^3 - 4x.$$
$$U_4(x) = 2x(8x^3 - 4x) - (4x^2 - 1) = 16x^4 - 12x^2 + 1.$$

Takeaways 4.3

- The first-kind and second-kind Chebyshev polynomials use the same recurrence form.
- The starting values determine which family is produced.
- Organize recurrence work line by line to avoid algebra slips.

Problem 4.12: Alternative telescoping patterns

(i) Evaluate $\sum_{n=0}^{4}(-1)^{n+1}(n+5)$.

(ii) Show $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$.

(iii) Hence find

$$S_N = \sum_{n=1}^{N}\left(\frac{1}{n} - \frac{1}{n+1}\right)$$

and $\lim_{N\to\infty} S_N$.

Hint: Part (iii) is a direct telescoping sum after part (ii).

Solution 4.12

$$\sum_{n=0}^{4}(-1)^{n+1}(n+5) = -7.$$

Also

$$\frac{1}{n(n+1)} = \frac{(n+1)-n}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

So

$$S_N = 1 - \frac{1}{N+1}, \qquad \lim_{N\to\infty} S_N = 1.$$

Problem 4.13: Surd telescoping

Let

$$a_k = \frac{1}{\sqrt{k+1}+\sqrt{k}}, \qquad S_n = \sum_{k=1}^{n} a_k.$$

(i) Show $a_k = \sqrt{k+1} - \sqrt{k}$.

(ii) Find S_n.

(iii) Evaluate S_{99}.

(iv) Find the smallest integer N such that $S_N > 10$.

Hint: Rationalize with the conjugate and then telescope.

Solution 4.13

$$a_k = \frac{\sqrt{k+1}-\sqrt{k}}{(k+1)-k} = \sqrt{k+1} - \sqrt{k}.$$

Hence

$$S_n = (\sqrt{2}-\sqrt{1}) + \cdots + (\sqrt{n+1}-\sqrt{n}) = \sqrt{n+1} - 1.$$

So $S_{99} = 10 - 1 = 9$. For $S_N > 10$:

$$\sqrt{N+1} - 1 > 10 \implies N > 120,$$

smallest such integer is $N = 121$.

Problem 4.14: Central symmetry in a GP

In a GP, define $P_n = T_1T_2\cdots T_n$.

(i) Show $P_n = a^n r^{\frac{n(n-1)}{2}}$.

(ii) Show $T_kT_{n-k+1} = T_1T_n$.

(iii) If $n = 11$ and middle term $T_6 = 5$, find P_{11}.

Hint: Use $T_k = ar^{k-1}$ and pair symmetric terms.

Solution 4.14

$$P_n = a(ar)\cdots(ar^{n-1}) = a^n r^{0+1+\cdots+(n-1)} = a^n r^{\frac{n(n-1)}{2}}.$$

Also

$$T_k T_{n-k+1} = ar^{k-1}\cdot ar^{n-k} = a^2 r^{n-1} = T_1 T_n.$$

For odd $n = 11$, the product is middle-term power:

$$P_{11} = T_6^{11} = 5^{11}.$$

Problem 4.15: Fibonacci and Lucas

Define

$$F_1 = 1,\ F_2 = 1,\ F_n = F_{n-1} + F_{n-2},$$

$$L_1 = 1,\ L_2 = 3,\ L_n = L_{n-1} + L_{n-2}.$$

Show that

$$L_n = F_{n-1} + F_{n+1}.$$

Hint: Set $A_n = L_n + F_n$ and $B_n = L_n - F_n$, then compare with Fibonacci terms.

Solution 4.15

From initial terms and recurrence,

$$A_n = L_n + F_n = 2F_{n+1}, \qquad B_n = L_n - F_n = 2F_{n-1}.$$

Adding gives

$$2L_n = 2F_{n+1} + 2F_{n-1},$$

so

$$L_n = F_{n-1} + F_{n+1}.$$

Problem 4.16: Linear combinations of two GPs

Let $T_n = ar^{n-1}$ and $U_n = AR^{n-1}$. Define $W_n = T_n + U_n$.

(i) Show (W_n) satisfies

$$W_{n+2} - (r+R)W_{n+1} + rRW_n = 0.$$

(ii) Solve

$$X_{n+2} - 5X_{n+1} + 6X_n = 0, \quad X_1 = 5,\ X_2 = 13.$$

Hint: For (ii), the characteristic roots are 2 and 3.

Solution 4.16

Substitute $W_n = ar^{n-1} + AR^{n-1}$ directly to verify (i). For (ii), write

$$X_n = \alpha 2^{n-1} + \beta 3^{n-1}.$$

Using $X_1 = 5,\ X_2 = 13$:

$$\alpha + \beta = 5, \quad 2\alpha + 3\beta = 13$$

gives $\alpha = 2,\ \beta = 3$. Hence

$$X_n = 2^n + 3^n.$$

4.3 Advanced Sequence Problems

Problem 4.17: Convergence condition

Consider the geometric series

$$\sum_{n=0}^{\infty} 7\left(\frac{m-1}{3}\right)^n.$$

Find all real values of m for which the series converges, and then state its sum.

Hint: Use the condition $|r| < 1$ for convergence of an infinite GP.

Solution 4.17

Here

$$r = \frac{m-1}{3}.$$

Convergence requires

$$\left|\frac{m-1}{3}\right| < 1 \implies |m-1| < 3 \implies -2 < m < 4.$$

For these values,

$$S_\infty = \frac{7}{1-\frac{m-1}{3}} = \frac{21}{4-m}.$$

Problem 4.18: AP and GP crossover

A sequence is both arithmetic and geometric. Given $a_1 = 6$ and $a_3 = 6$, find all possible sequences.

Hint: Let AP difference be d and GP ratio be r. Use both models on a_3.

Solution 4.18

AP gives

$$a_3 = a_1 + 2d = 6 + 2d = 6 \implies d = 0,$$

so AP is constant: $a_n = 6$. For GP,

$$a_3 = a_1 r^2 = 6r^2 = 6 \implies r^2 = 1 \implies r = \pm 1.$$

If $r = -1$, terms alternate $6, -6, 6, \ldots$ and are not arithmetic. Hence the only sequence that is both AP and GP is

$$a_n = 6 \quad \text{for all } n,$$

with $d = 0$ and $r = 1$.

Problem 4.19: Chebyshev roots

The first-kind Chebyshev polynomial of degree 5 is

$$T_5(x) = 16x^5 - 20x^3 + 5x.$$

Use the identity $T_5(\cos\theta) = \cos(5\theta)$ to find all roots of $T_5(x)$ in the interval $[-1, 1]$.

Hint: Set $\cos(5\theta) = 0$, then convert the resulting angles back to $x = \cos\theta$.

Solution 4.19

Let $x = \cos\theta$. We need

$$\cos(5\theta) = 0.$$

Thus

$$5\theta = \frac{(2k+1)\pi}{2}, \qquad k = 0, 1, 2, 3, 4.$$

So

$$\theta = \frac{(2k+1)\pi}{10},$$

and the five roots are

$$x = \cos\frac{\pi}{10},\ \cos\frac{3\pi}{10},\ \cos\frac{5\pi}{10},\ \cos\frac{7\pi}{10},\ \cos\frac{9\pi}{10}.$$

Takeaways 4.4

- Roots of $T_n(x)$ on $[-1, 1]$ come from zeros of $\cos(n\theta)$.
- The substitution $x = \cos\theta$ is natural because $\cos\theta$ covers $[-1, 1]$.
- Degree 5 gives five roots, which is a useful check.

Problem 4.20: Architecture of GP sums

Let a GP have first term $a \neq 0$, ratio $r \neq 0, \pm 1$, and partial sums S_n.

(i) Prove $\dfrac{S_{2n}}{S_n} = r^n + 1$.

(ii) Given $S_{12} = 65 S_6$ and second term $ar = 12$, find possible (a, r).

(iii) If Σ_n is the partial-sum sequence of a GP with first term a and ratio r^2, show

$$\frac{\Sigma_n}{S_n} = \frac{r^n + 1}{r + 1}.$$

Hint: Use $S_n = \dfrac{a(r^n - 1)}{r - 1}$ and difference-of-squares factorization.

Solution 4.20

$$\frac{S_{2n}}{S_n} = \frac{r^{2n} - 1}{r^n - 1} = r^n + 1.$$

So $r^6 + 1 = 65 \implies r^6 = 64 \implies r = \pm 2$. Since $ar = 12$, possible pairs are $(a, r) = (6, 2)$ or $(-6, -2)$. Also

$$\Sigma_n = \frac{a(r^{2n} - 1)}{r^2 - 1},$$

thus

$$\frac{\Sigma_n}{S_n} = \frac{r^{2n} - 1}{(r + 1)(r^n - 1)} = \frac{r^n + 1}{r + 1}.$$

Problem 4.21: AP space and difference operator

Let $\mathcal{A}(a, d)$ denote an AP. Define

$$\mathcal{I} = \mathcal{A}(1, 0), \qquad \mathcal{J} = \mathcal{A}(0, 1).$$

(i) Show any AP is uniquely $a\mathcal{I} + d\mathcal{J}$.

(ii) Let $\Delta x_n = x_{n+1} - x_n$. Show $\Delta \mathcal{I} = 0$ and $\Delta \mathcal{J} = \mathcal{I}$.

(iii) Deduce $\Delta(a\mathcal{I} + d\mathcal{J}) = d\mathcal{I}$.

Hint: Compare first term and common difference.

Solution 4.21

An AP has nth term $a + (n-1)d = a \cdot 1 + d(n-1)$, so $X = a\mathcal{I} + d\mathcal{J}$, unique by matching (a, d). Also $\mathcal{I} = (1, 1, 1, \dots)$ so $\Delta \mathcal{I} = 0$, and $\mathcal{J} = (0, 1, 2, \dots)$ so $\Delta \mathcal{J} = (1, 1, 1, \dots) = \mathcal{I}$. Hence

$$\Delta(a\mathcal{I} + d\mathcal{J}) = a\Delta\mathcal{I} + d\Delta\mathcal{J} = d\mathcal{I}.$$

Problem 4.22: Basis and projection in AP space

Let $\mathcal{E}_1 = \mathcal{A}(1,1)$ and $\mathcal{E}_2 = \mathcal{A}(1,-1)$.

(i) Express any AP $\mathcal{A}(a,d)$ as $\lambda\mathcal{E}_1 + \mu\mathcal{E}_2$.

(ii) Decompose $P = (5, 8, 11, 14, \dots)$ in this basis.

Hint: Match first terms and differences: $\lambda + \mu = a$, $\lambda - \mu = d$.

Solution 4.22

Solving

$$\lambda + \mu = a, \qquad \lambda - \mu = d$$

gives

$$\lambda = \frac{a+d}{2}, \qquad \mu = \frac{a-d}{2}.$$

For P, $a = 5$, $d = 3$, so $\lambda = 4$, $\mu = 1$.

Problem 4.23: Countable and uncountable sets

(i) Explain why positive rationals are countable by diagonal listing.

(ii) Use Cantor's diagonal argument to show reals in $[0, 1)$ are uncountable.

Hint: Construct a new decimal differing from the nth listed decimal at position n.

Solution 4.23

Rationals can be arranged in an infinite grid $\frac{p}{q}$ and traversed by diagonals, skipping duplicates after simplification; this yields a sequence listing all positive rationals. For reals in $[0, 1)$, assume a list $T_1, T_2, \dots$ exists. Build x so its nth digit differs from the nth digit of T_n. Then $x \neq T_n$ for every n, contradicting completeness of the list. Hence uncountable.

Problem 4.24: Prime denominators and cycle length

Let p be prime, $p \neq 2, 5$, and L be period length of $\frac{1}{p}$.

(i) Show L is the least positive integer with $10^L \equiv 1 \pmod{p}$.

(ii) Deduce $L \mid (p-1)$.

(iii) Find periods for $\frac{1}{37}$ and $\frac{1}{41}$.

Hint: Use Fermat's theorem and first occurrence of $p \mid (10^n - 1)$.

Solution 4.24

Recurring block theory gives $(10^L - 1)/p \in \mathbb{Z}$, hence $10^L \equiv 1 \pmod p$; minimality gives least such L. By Fermat, $10^{p-1} \equiv 1 \pmod p$, so the order L divides $p-1$. Since $37 \mid (10^3 - 1)$ and not $10^1 - 1, 10^2 - 1$, period is 3. Since $41 \mid (10^5 - 1)$ and not $10^1 - 1$, period is 5.

Problem 4.25: Pythagorean means

For $0 < a < b$, define

$$A = \frac{a+b}{2}, \quad G = \sqrt{ab}, \quad H = \frac{2ab}{a+b}.$$

(i) Prove $H < G < A$.

(ii) Find $\frac{b}{a}$ such that a, G, A is an AP.

(iii) Show H, G, A cannot be an AP unless $a = b$.

Hint: Use $(\sqrt{a} - \sqrt{b})^2 > 0$ and $G^2 = AH$.

Solution 4.25

From $(\sqrt{a} - \sqrt{b})^2 > 0$, get $A > G$. Also

$$AH = \frac{a+b}{2} \cdot \frac{2ab}{a+b} = ab = G^2,$$

so H, G, A form a GP and therefore $H < G < A$. For a, G, A in AP:

$$2G = a + A \implies 4\sqrt{ab} = 3a + b.$$

Let $x = b/a$; then $(x-9)(x-1) = 0$, and $x > 1$ gives $x = 9$. If H, G, A were AP, then $2G = H + A = G^2/A + A$, forcing $A = G$, contradiction for $a \neq b$.

Problem 4.26: Recurring decimals and modular structure

Let p be prime ($p \neq 2, 5$), and let n be the period of $\frac{1}{p}$.

(i) Show $n \mid (p-1)$.

(ii) If $n = 2k$, prove $10^k \equiv -1 \pmod p$ and the two half-blocks of the repetend sum to $10^k - 1$.

Hint: Factor $10^{2k} - 1 = (10^k - 1)(10^k + 1)$ and use minimality of n.

Solution 4.26

From order arguments, n is the multiplicative order of 10 mod p, so $n \mid (p-1)$. If $n = 2k$, then $p \mid (10^{2k} - 1)$. Since $2k$ is minimal, $p \nmid (10^k - 1)$, hence $p \mid (10^k + 1)$, so $10^k \equiv -1 \pmod p$. Writing repetend as two k-digit blocks A, B gives Midy's theorem:

$$A + B = 10^k - 1.$$

Problem 4.27: Sissa and arithmetico-geometric sums

(i) Evaluate $S_{64} = 1 + 2 + 4 + \cdots + 2^{63}$.

(ii) Let
$$T = \sum_{n=1}^{64} n2^{n-1}.$$
Use shift-and-subtract to find a closed form.

(iii) Evaluate
$$\sum_{n=1}^{\infty} \frac{n}{2^n}.$$

Hint: Use $T - 2T$ for the finite AGP, and similarly $S - \frac{1}{2}S$ for the infinite one.

Solution 4.27

$$S_{64} = 2^{64} - 1.$$

Also
$$T - 2T = \sum_{n=1}^{64} 2^{n-1} - 64 \cdot 2^{64} = (2^{64} - 1) - 64 \cdot 2^{64},$$
so
$$T = 63 \cdot 2^{64} + 1.$$
For $S = \sum_{n=1}^{\infty} \frac{n}{2^n}$:
$$S - \frac{1}{2}S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1,$$
hence $S = 2$.

Problem 4.28: Discrete calculus and sums of powers

∇ is the backward difference operator, defined as $\nabla f(n) = f(n) - f(n-1)$ for discrete functions $f(n)$. In this problem, you will explore the properties of the backward difference operator and its applications to sums of powers.
For example, The first order of ∇ is

$$\nabla(n^2) = n^2 - (n-1)^2 = 2n - 1$$

.
The second order of ∇ is

$$\nabla^2 f(n) = \nabla(\nabla f(n)) = \nabla(f(n) - f(n-1)) = f(n) - 2f(n-1) + f(n-2)$$

and the third order of ∇ is

$$\nabla^3 f(n) = \nabla(\nabla^2 f(n)) = \nabla(f(n) - 2f(n-1) + f(n-2)) = f(n) - 3f(n-1) + 3f(n-2) - f(n-3)$$

.
In this problem, we use the ∇ operator to derive formulas for sums of powers.

(i) Show $n^3 - (n-1)^3 = 3n^2 - 3n + 1$.

(ii) Sum this identity from 1 to N to derive

$$\sum_{n=1}^{N} n^2 = \frac{N(N+1)(2N+1)}{6}.$$

(iii) Evaluate $\nabla^3(n^3)$

(iv) State the leading term of $\sum_{n=1}^{N} n^k$.

Hint: Use telescoping on the left and known formulas for $\sum n$.

Solution 4.28

Expansion gives part (i). Summing:

$$N^3 = 3\sum_{n=1}^{N} n^2 - 3\sum_{n=1}^{N} n + N,$$

which simplifies to

$$\sum_{n=1}^{N} n^2 = \frac{N(N+1)(2N+1)}{6}.$$

Also $\nabla^3(n^3) = 6$. In general,

$$\sum_{n=1}^{N} n^k = \frac{1}{k+1} N^{k+1} + (\text{lower powers of } N).$$

Problem 4.29: The Fibonacci Generating Function

Let F_n be the n-th Fibonacci number ($F_1 = 1, F_2 = 1, F_n = F_{n-1} + F_{n-2}$). We define the power series $S(x)$ as:

$$S(x) = \sum_{n=1}^{\infty} F_n x^{n+1}$$

(i) By expanding $S(x) - xS(x) - x^2 S(x)$ and using the recurrence property of Fibonacci numbers, prove that:

$$S(x) = \frac{x^2}{1 - x - x^2}$$

(ii) Given the growth rate of Fibonacci numbers, explain why this series converges for $x = 0.1$.

(iii) Show that $S(0.1) = \frac{1}{89}$.

(iv) The decimal expansion of $\frac{1}{89}$ is $0.011235955\ldots$. Explain why the digit in the 10^{-7} place is 9 rather than the expected Fibonacci number $F_6 = 8$.

Hint: Consider the definition of $S(x)$ term by term for (i). For (iv), look closely at how the next Fibonacci number $F_7 = 13$ interacts with the decimal places.

Solution 4.29

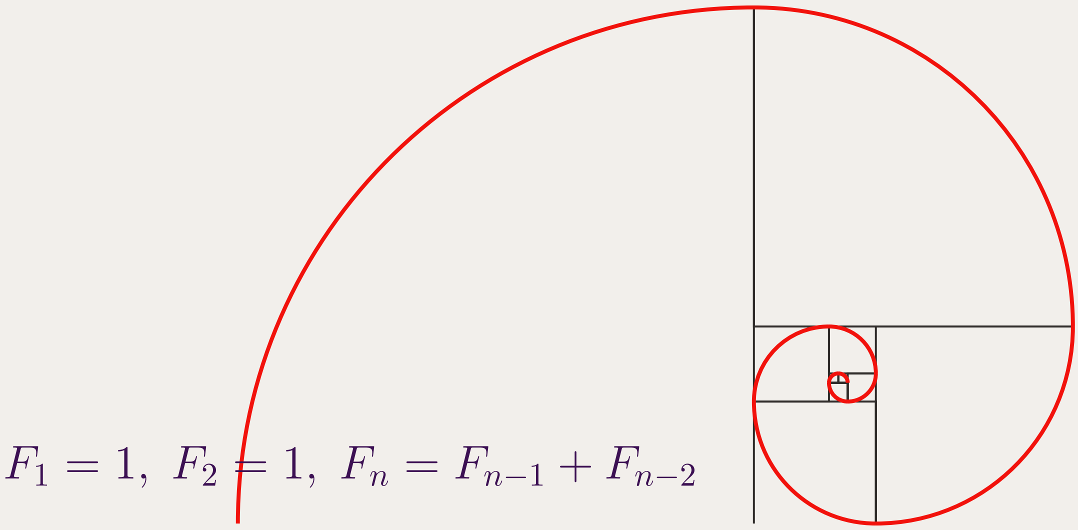
(i) Substitution into $S(x)(1 - x - x^2)$ cancels out all terms except the first, leaving $F_1 x^2 = x^2$. Hence $S(x) = \frac{x^2}{1-x-x^2}$.
(ii) Fibonacci numbers grow roughly geometrically by the golden ratio $\phi \approx 1.618$. Because $1.618\cdots \times 0.1 < 1$, the series converges.
(iii) Substitute $x = 0.1$:

$$S(0.1) = \frac{0.1^2}{1 - 0.1 - 0.1^2} = \frac{0.01}{1 - 0.1 - 0.01} = \frac{0.01}{0.89} = \frac{1}{89}.$$

(iv) The required sum gives $F_6 \times (0.1)^7 + F_7 \times (0.1)^8$. $F_6 = 8$ and $F_7 = 13$, so the sum overlapping the 10^{-7} place includes the carry-over from F_7: $8 + 1 = 9$.

Takeaways 4.5

- Power series generating functions can be explicitly derived using recurrence relations.
- The decimal expansion of fractions naturally encodes the base-10 carry-over of power series evaluations.
- Fibonacci recursion can also be visualised geometrically via a quarter-turn spiral whose radii follow consecutive Fibonacci numbers.



5 Conclusion

Strong sequence work in Extension 1 is not just memorizing formulas. It is about choosing the right model, controlling notation, and checking whether each step is logically justified.

The key habits are:

- define terms clearly, including first term and common difference or ratio
- check indexing before and after every sigma manipulation
- identify whether a question is about terms, partial sums, or limits
- test answers for reasonableness by checking growth and sign behavior

With steady practice, sequences become one of the most reliable scoring areas in Extension 1.

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6 Appendices

6.1 Appendix A: Formula Sheet

- Arithmetic nth term:
$$a_n = a + (n-1)d.$$
- Arithmetic sum of first n terms:
$$S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l).$$
- Geometric nth term:
$$a_n = ar^{n-1}.$$
- Geometric sum of first n terms ($r \neq 1$):
$$S_n = a\frac{1-r^n}{1-r} = a\frac{r^n - 1}{r-1}.$$
- Infinite geometric sum ($|r| < 1$):
$$S_\infty = \frac{a}{1-r}.$$
- Sums of powers for the first n positive integers:
$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2},$$
$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6},$$
$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

Chebyshev Polynomials

The Chebyshev polynomials of the first kind, denoted $T_n(x)$, are defined by
$$T_0(x) = 1, \qquad T_1(x) = x, \qquad T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \quad (n \geq 1).$$
Equivalently,
$$T_n(\cos\theta) = \cos(n\theta).$$
The Chebyshev polynomials of the second kind, denoted $U_n(x)$, are defined by
$$U_0(x) = 1, \qquad U_1(x) = 2x, \qquad U_{n+1}(x) = 2xU_n(x) - U_{n-1}(x) \quad (n \geq 1).$$
Equivalently,
$$U_n(\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta} \quad (\sin\theta \neq 0).$$

Degree	First kind $T_n(x)$	Second kind $U_n(x)$
1	x	$2x$
2	$2x^2 - 1$	$4x^2 - 1$
3	$4x^3 - 3x$	$8x^3 - 4x$
4	$8x^4 - 8x^2 + 1$	$16x^4 - 12x^2 + 1$
5	$16x^5 - 20x^3 + 5x$	$32x^5 - 32x^3 + 6x$

6.2 Appendix B: Common Sequence Traps

- Mixing up term formulas and sum formulas: a_n and S_n answer different questions.
- Losing track of index starts: check whether the sequence starts at $n = 0$ or $n = 1$.
- Using $S_\infty = \frac{a}{1-r}$ when $|r| \geq 1$: this formula only applies for $|r| < 1$.
- Sign errors in GP sums: keep the exact formula form and substitute carefully.
- Assuming recursion converges without proof: identify monotonicity or contraction before claiming limits.
- Treating recurring decimals as finite truncations: they are infinite geometric expansions.

6.3 Appendix C: Rigorous Definition of the Limit of a Sequence

Definition 1 (ϵ-N Definition of a Limit)**.** Let (a_n) be a sequence of real numbers. We say that (a_n) **converges to a limit** L, written

$$\lim_{n\to\infty} a_n = L,$$

if for every real number $\epsilon > 0$ there exists a positive integer N such that

$$n > N \implies |a_n - L| < \epsilon.$$

In formal logical notation:

$$\forall\,\epsilon > 0,\ \exists\,N \in \mathbb{N} \text{ such that } n > N \implies |a_n - L| < \epsilon.$$

If no such L exists, the sequence is said to **diverge**.

Unpacking the definition.

- $\epsilon > 0$ **is the "margin of error."** The challenger picks any strictly positive number, no matter how small. A smaller ϵ places a tighter band around L.
- N **is the "cutoff index."** The responder must produce a concrete N that works for the chosen ϵ. Typically, smaller ϵ forces a larger N.
- **Only the *tail* matters.** The condition $n > N$ means the first N terms may behave arbitrarily; only the infinite tail of the sequence is constrained.
- $|a_n - L| < \epsilon$ **is the "closeness" condition.** Every term beyond index N must lie strictly inside the interval $(L - \epsilon,\ L + \epsilon)$.

Visual interpretation (the "ϵ-tube").

Plot the sequence on a graph with the index n on the horizontal axis and a_n on the vertical axis. Draw a horizontal dashed line at $y = L$. Choosing $\epsilon > 0$ creates a horizontal band $\big(L-\epsilon,\ L+\epsilon\big)$ of total width 2ϵ around the limit. The definition states that for every such band there is a vertical line $x = N$ such that *every point of the sequence to the right of that line lies inside the band.*

Example: Proving $\lim\limits_{n\to\infty} \dfrac{1}{n} = 0$**.**

Proof. Let $\epsilon > 0$ be given.

Scratchwork. We require

$$\left|\frac{1}{n} - 0\right| < \epsilon.$$

Since $n \in \mathbb{N}$ is positive, this simplifies to $\frac{1}{n} < \epsilon$, i.e. $n > \frac{1}{\epsilon}$.

Choice of N. Let N be any positive integer satisfying $N > \frac{1}{\epsilon}$. (Such an integer exists by the Archimedean property of $\mathbb{R}$.)

Verification. Suppose $n > N$. Then

$$n > N > \frac{1}{\epsilon} \implies \frac{1}{n} < \epsilon \implies \left|\frac{1}{n} - 0\right| = \frac{1}{n} < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, the definition is satisfied and $\lim_{n\to\infty} \frac{1}{n} = 0$. □

Concrete illustration.

For $\epsilon = 0.01$ we need $N > 100$, so we may take $N = 101$. Every term $a_n = \frac{1}{n}$ with $n \geq 102$ satisfies $\frac{1}{102} < 0.01$, placing all those terms inside the band $(-0.01, 0.01)$.

Note: The following advanced theorems are outside the scope of the regular HSC syllabus, but they are extremely good-to-know fundamentals for higher education and university mathematics.

1. Bolzano-Weierstrass Theorem

The Bolzano-Weierstrass Theorem is a "guarantee of stability." It tells us that as long as a sequence doesn't head off to infinity, it must contain at least one "hidden" convergent behavior.

- **The Intuition:** Imagine a sequence as a set of points trapped inside a finite box (bounded). Even if the points bounce around randomly and never settle on one spot as a whole (like $a_n = (-1)^n$), the points must eventually start "clustering" somewhere because they have nowhere else to go.
- **Subsequences:** A subsequence is formed by picking infinitely many terms from the original sequence in their original order.
  - *Example:* Take $a_n = (-1)^n \to \{-1, 1, -1, 1, \dots\}$. The sequence itself oscillates and diverges. However, if we pick only the even terms, we get $\{1, 1, 1, \dots\}$, which converges to 1.
- **The Utility:** In proofs, if you know a sequence is bounded, you can immediately "extract" a convergent subsequence to use for further logical steps.

2. Cauchy Criterion for Convergence

The Cauchy Criterion is the ultimate "internal" test for convergence.

- **The Definition:** In a standard limit, we say terms get close to a **limit** L. In a Cauchy sequence, we say the terms get **close to each other**.

- **Formal Statement:** For every $\epsilon > 0$, there exists an N such that for all $m, n > N$:

$$|a_n - a_m| < \epsilon$$

- **Why it Matters:** Usually, to prove a sequence converges, you first have to *guess* what the limit L is. The Cauchy Criterion removes this requirement. If you can prove that the terms of the sequence eventually "huddle together" and the gaps between them shrink to zero, the sequence **must** converge to some real number.

- **The "Completeness" of Real Numbers:** This theorem works because the Real Number line has no "holes." If the terms are huddling together, they are guaranteed to be huddling around a real number. (This is not true in the Rational numbers, where a sequence of fractions could huddle around an "irrational hole" like $\sqrt{2}$).

6.4 Appendix D: Limit Superior and Limit Inferior

Note on Syllabus

The concepts of limit superior (lim sup) and limit inferior (lim inf) are strictly **outside the scope** of the HSC Mathematics Extension 1 and Extension 2 syllabuses. However, they provide an excellent bridge into higher mathematics (such as university-level real analysis) and serve as useful tools to "estimate a boundary" for the limits of sequences and functions.

Estimating Boundaries with Limsup and Liminf

In standard mathematics, the limit of a sequence a_n as $n \to \infty$ describes the exact value the sequence settles on. However, many sequences do not settle on a single value; they might oscillate or jump around (for example, $a_n = (-1)^n$).

Instead of asking "What is the exact limit?", we can ask "What are the ultimate upper and lower boundaries of this sequence?"

- **Limit Superior ($\limsup_{n\to\infty} a_n$):** The eventual upper bound of the sequence. It describes the maximum value that the sequence continues to reach or approach as n becomes arbitrarily large.

- **Limit Inferior ($\liminf_{n\to\infty} a_n$):** The eventual lower bound of the sequence. It describes the minimum value that the sequence continues to reach or approach as n becomes arbitrarily large.

A sequence (a_n) converges to a limit L if and only if both its limit superior and limit inferior are equal to that same expected limit L. That is:

$$\lim_{n\to\infty} a_n = L \iff \limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n = L$$

Simple Examples

Example 1: A Converging Sequence

Consider the sequence $a_n = \frac{1}{n}$. The terms are $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$ As $n \to \infty$, the sequence approaches 0. Here, the eventual upper boundary and lower boundary are the same.

$$\limsup_{n\to\infty} \frac{1}{n} = 0 \quad \text{and} \quad \liminf_{n\to\infty} \frac{1}{n} = 0$$

Since they are equal, the standard limit exists and $\lim_{n\to\infty} \frac{1}{n} = 0$.

Example 2: An Oscillating Sequence

Consider the sequence $a_n = (-1)^n$. The terms are $-1, 1, -1, 1, -1, 1, \ldots$ This sequence never settles on a single converging value, so the standard limit does not exist. However, the sequence is clearly bounded.

- The highest value the sequence keeps hitting as $n \to \infty$ is 1. Thus, $\limsup_{n\to\infty}(-1)^n = 1$.
- The lowest value the sequence keeps hitting as $n \to \infty$ is -1. Thus, $\liminf_{n\to\infty}(-1)^n = -1$.

Example 3: A Sequence with Growth and Oscillation

Consider $a_n = 5 + \frac{(-1)^n}{n}$. The sequence oscillates slightly above and below 5, getting closer and closer to 5.

$$\limsup_{n\to\infty}\left(5 + \frac{(-1)^n}{n}\right) = 5 \quad \text{and} \quad \liminf_{n\to\infty}\left(5 + \frac{(-1)^n}{n}\right) = 5$$

Both estimates boundary conditions pinch together at 5, confirming the exact limit is 5.

6.5 Appendix E: Discrete Calculus and the Difference Operator

Note on Syllabus

The concept of the backward difference operator (∇) belongs to the field of *Discrete Calculus*. While the mechanics of telescoping sums are implicitly part of Extension 1 and Extension 2, formalizing them into discrete derivative operators is **outside the scope** of the formal HSC syllabus. Nevertheless, it deeply connects the study of sequences to continuous calculus that is taught at the university level.

The Difference Operator: The "Derivative" for Sequences

In continuous calculus, the derivative measures the instantaneous rate of change of a function $f(x)$ with respect to x:

$$f'(x) = \lim_{h\to 0} \frac{f(x) - f(x-h)}{h}$$

When we study sequences, the domain is restricted to discrete integers. This means the smallest possible change or "step" in the variable n is exactly 1 (we cannot let $h \to 0$). Therefore, the discrete counterpart to the derivative is the **backward difference operator**, denoted by ∇ (nabla):

$$\nabla f(n) = \frac{f(n) - f(n-1)}{1} = f(n) - f(n-1)$$

(Note: There is also a forward difference operator $\Delta f(n) = f(n+1) - f(n)$, which functions very similarly).

Just as we can take multiple derivatives of functions like $f''(x)$, we can repeatedly apply difference operators to sequences, evaluating things like $\nabla^3(n^3)$.

Why is it Useful?

In calculus, the Fundamental Theorem of Calculus links derivatives to integrals:

$$\int_a^b f'(x)\,dx = f(b) - f(a)$$

In discrete mathematics involving sequences, the operation equivalent to continuous integration is **summation**. Summing a difference operator perfectly mirrors the Fundamental Theorem of Calculus through **telescoping sums**:

$$\sum_{n=1}^{N} \nabla f(n) = \sum_{n=1}^{N} (f(n) - f(n-1)) = f(N) - f(0)$$

This elegant connection turns difficult summations into basic boundary evaluations. If you want to compute the sum of a sequence $g(n)$, and you can guess or find a discrete "anti-derivative" $f(n)$ such that $\nabla f(n) = g(n)$, the sum is trivially computed.

Application to Sums of Powers

Consider the continuous derivative of a polynomial power: $\frac{d}{dx}(x^3) \approx 3x^2$. In the discrete world, the difference operator applied to n^3 yields:

$$\nabla(n^3) = n^3 - (n-1)^3 = 3n^2 - 3n + 1$$

By summing this identity, we mimic the concept of integrating a derivative:

$$\sum_{n=1}^{N} \nabla(n^3) = \sum_{n=1}^{N} (3n^2 - 3n + 1)$$

Because the left hand side is just a sum of differences, it telescopes and collapses, leaving only the bounds:

$$N^3 - 0^3 = 3\sum_{n=1}^{N} n^2 - 3\sum_{n=1}^{N} n + \sum_{n=1}^{N} 1$$

This equation is immensely powerful: it provides a rapid mechanism to find the closed-form formula for $\sum_{n=1}^{N} n^2$. Furthermore, it intuitively explains why the continuous integral $\int x^2\, dx = \frac{x^3}{3}$ corresponds directly to the discrete sum having a leading polynomial term of $\frac{N^3}{3}$.

